DIGITAL LOGIC DESIGN / PROJECT REPORT

4-BIT BINARY TO DECIMAL CONVERSION AND DISPLAY SYSTEM

BCD decoding • Boolean logic • Seven-segment display

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Submitted as part of the Digital Logic Design project

A concise technical report on converting a four-bit input into a readable decimal digit using logic gates and a seven-segment display.

01 Abstract

This project presents a four-bit input system intended to display decimal digits from 0 to 9 on a seven-segment display. The input bits B3, B2, B1 and B0 carry weights 8, 4, 2 and 1. Truth-table analysis and Boolean logic are used to determine which display segments should turn ON for each valid digit. The report introduces AND, OR and NOT gates, Karnaugh-map (K-map) simplification, and a BCD-to-seven-segment decoder such as the IC 7447.

Keywords: Binary input, BCD, Boolean algebra, logic gates, Karnaugh map, seven-segment display.

02 Introduction

Digital systems store and process information using binary values, 0 and 1. People commonly read numbers in decimal form. A conversion and display circuit bridges this difference by accepting a four-bit input and producing the segment pattern required to show the corresponding digit.

A four-bit input has 16 possible combinations, from 0000 to 1111. In 8421 Binary-Coded Decimal (BCD), only 0000 through 1001 represent the decimal digits 0 through 9. The remaining combinations, 1010 through 1111, are invalid for a single BCD digit and may be treated as don't-care states during minimization when the design permits it.

03 Problem Statement & Objectives

> **Problem statement:** Design a logic system that accepts a four-bit binary-coded input and displays the corresponding decimal digit from 0 to 9 using a seven-segment display.

- Understand the relationship between four-bit binary input and decimal digits.
- Prepare a truth table for the valid input range 0000–1001.
- Develop Boolean expressions for display segments a through g.
- Represent the required logic using AND, OR and NOT operations.
- Explain the connection to a seven-segment display or a suitable BCD decoder.

04 Components & Design Elements

Element	Purpose
Four input bits (B3–B0)	Represent the input value using 8421 weights.
AND gates	Produce 1 when all required inputs are 1.
OR gates	Produce 1 when at least one input is 1.
NOT gates	Invert an input: 0 becomes 1 and 1 becomes 0.
Karnaugh map (K-map)	Simplifies Boolean expressions and can reduce gate requirements.
BCD-to-seven-segment decoder (e.g., IC 7447)	Converts BCD inputs into signals for the display segments.
Seven-segment display	Shows decimal digits by lighting selected segments.

05 Theory & Concepts

5.1 Four-bit binary input

The input bits are written as B3 B2 B1 B0 and have weights 8, 4, 2 and 1. The represented value is:

Value = 8B3 + 4B2 + 2B1 + B0

Example: 0111 = (0 × 8) + (1 × 4) + (1 × 2) + (1 × 1) = 7.

5.2 Binary-Coded Decimal (BCD)

BCD uses a four-bit code for each decimal digit. The valid codes for a single digit are 0000 (0) through 1001 (9). Codes 1010–1111 do not represent valid single decimal digits in 8421 BCD.

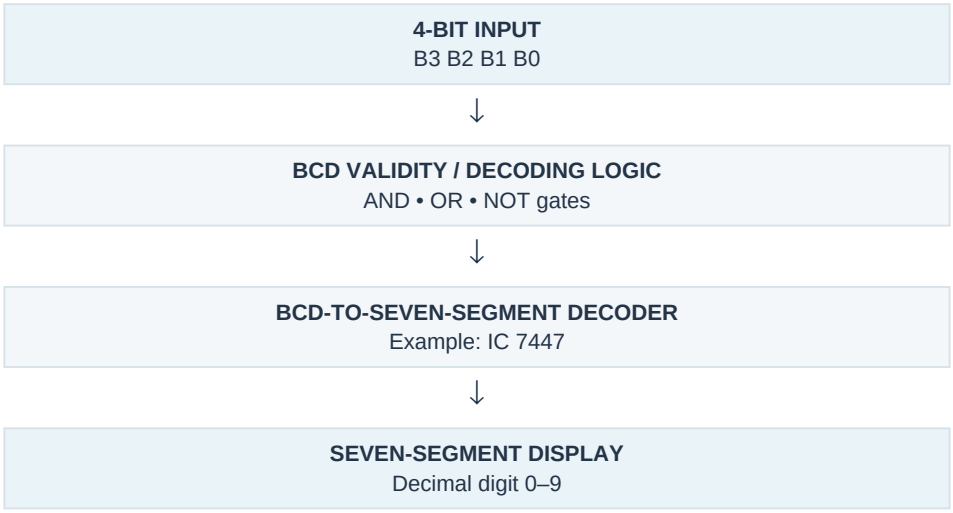
5.3 Seven-segment display

A seven-segment display has individually controlled segments labelled a, b, c, d, e, f and g. Turning selected segments ON and the others OFF forms a digit. In the active-high convention used in this report, 1 means ON and 0 means OFF; actual electrical polarity depends on the display and driver IC.

5.4 Logic operations

- AND (·): output is 1 only when all inputs are 1.
- OR (+): output is 1 when at least one input is 1.
- NOT (′): produces the complement of a bit; for example, B0′ is NOT B0.

06 System Block Diagram



Implementation note: The report depicts decoding logic followed by a BCD-to-seven-segment decoder. If the seven Boolean expressions directly drive the segments, the separate decoder stage may not be needed. The final circuit should use one consistent implementation.

07 Truth Table for Valid BCD Inputs

Decimal	B3	B2	B1	B0	Binary code
0	0	0	0	0	0000
1	0	0	0	1	0001
2	0	0	1	0	0010
3	0	0	1	1	0011
4	0	1	0	0	0100
5	0	1	0	1	0101
6	0	1	1	0	0110
7	0	1	1	1	0111
8	1	0	0	0	1000
9	1	0	0	1	1001

Invalid BCD combinations: 1010, 1011, 1100, 1101, 1110 and 1111. These are treated as don't-care (X) conditions during K-map simplification in the source design.

08 Boolean Expressions for Display Segments

The following minimized expressions are transcribed from the supplied project document. Here, ′ means NOT, · means AND and + means OR.

Segment	Boolean expression
a	B3 + B1 + (B2·B0) + (B2′·B0′)
b	B2′ + (B1′·B0′) + (B1·B0)
c	B1 + B0′ + B2
d	(B2′·B1′·B0) + (B1·B0′) + (B2·B1′) + B3
e	(B1′·B0) + (B2·B0)
f	B3 + (B2·B1′) + (B2·B0′) + (B1′·B0′)
g	B3 + (B2·B1′) + (B1·B0′)

Verification required: These equations have been transcribed from the supplied project document and have not been independently verified against every truth-table row. Before building or submitting a physical circuit, verify the equations and output polarity against the intended segment patterns and the selected driver IC datasheet.

09 Gate-Level Implementation

Each Boolean expression can be implemented by connecting logic gates in the required order. NOT gates create complemented inputs, AND gates combine terms, and OR gates combine those terms into the final segment output.

Example: segment a

$$a = B3 + B1 + (B2 \cdot B0) + (B2' \cdot B0')$$

- Use NOT gates to generate B2′ and B0′.
- Use one AND gate for B2 · B0.
- Use another AND gate for B2′ · B0′.
- OR the terms B3, B1, (B2 · B0) and (B2′ · B0′) to obtain output a.

Apply the same method to segments b, c, d, e, f and g using their respective input combinations. The circuit can be drawn with individual gates or simulated in digital-logic software.

10 Working Principle

Step	Stage	Description
01	Input	Four switches or input signals provide the binary code B3B2B1B0.
02	Interpret	The 8-4-2-1 weights identify the decimal value.
03	Decode	The valid BCD input is processed by the logic network or a BCD-to-seven-segment decoder.
04	Select segments	The logic determines which of the seven segments must turn ON.
05	Display	The lit segments form the corresponding decimal digit.

Example: Input 0111 represents decimal 7. In the segment convention used here, segments a, b and c should be ON and the other segments OFF.

11 Applications

Application	Use
Digital clocks and timers	Show counter values as readable digits.
Calculators	Present numeric results to users.
Digital counters	Display event counts, queue numbers or production counts.
Measurement instruments	Show values measured by counters and meters.
Elevator / floor indicators	Display a floor number from a digital signal.
Embedded systems	Provide a simple numeric interface for electronic devices.

12 Advantages & Limitations

Advantages

- Uses simple, fundamental digital logic operations.
- Truth tables make the required input-output behaviour clear.
- K-map simplification can reduce the number of gates and connections.
- A seven-segment display provides a compact, easy-to-read numeric output.

Limitations and design considerations

- The valid single-digit BCD range is 0–9; inputs 10–15 need defined handling if they can occur.
- The display type and driver IC determine output polarity and wiring requirements.
- A real circuit requires suitable current-limiting resistors and correct power connections, as specified by component datasheets.
- The segment equations should be verified before hardware implementation.

13 Conclusion

This project explains the digital-logic path from a four-bit input to a readable decimal display. It introduces 8421 BCD, valid input combinations, truth-table construction, Boolean expressions, K-map simplification, basic logic gates and seven-segment display operation. The design illustrates how logical conditions can be combined to control individual display segments and connects classroom concepts to practical devices such as clocks, counters, calculators and measurement instruments.

> The final implementation should use one consistent architecture—either direct segment logic or a BCD-to-seven-segment decoder—and should be tested against all valid inputs from 0 to 9.

14 References

• Project presentation: “Design of a 4-Bit Binary to Decimal Conversion and Display System Using Logic Gates,” provided by the project team.

• Digital Logic Design course notes covering Boolean algebra, truth tables, Karnaugh maps and seven-segment displays.

• Manufacturer datasheet for the selected BCD-to-seven-segment driver IC (for example, IC 7447), to be consulted for pin connections, output polarity and electrical limits.